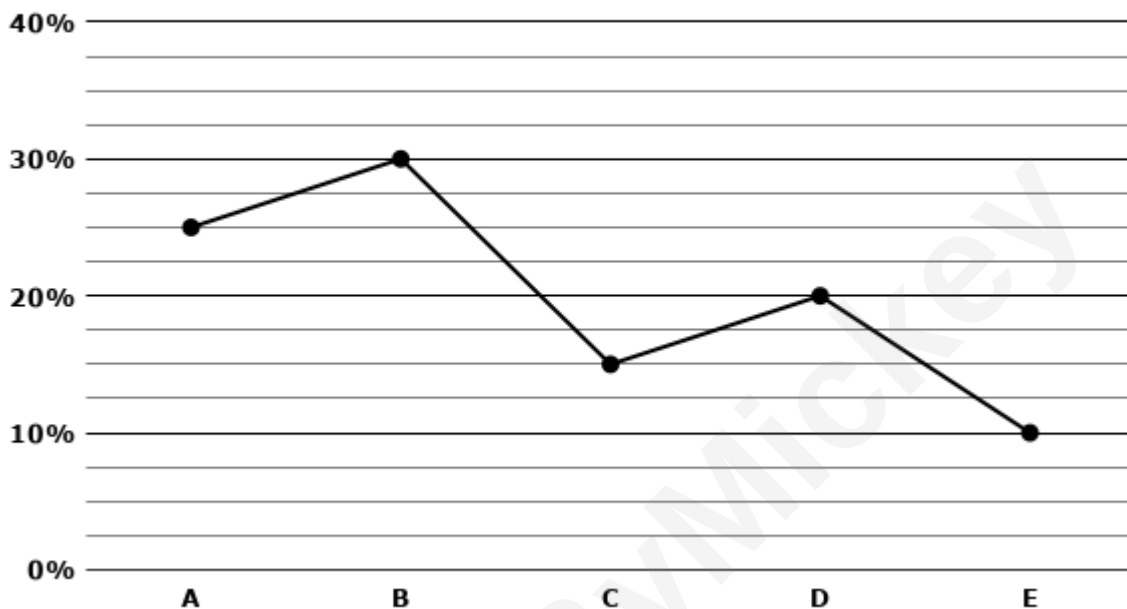


1. Questions

Study the following information carefully and answer the questions given below.

The line graph given below shows the percentage distribution of the number of keyboards sold by five different shops namely A, B, C, D and E.

The percentage distribution of the number of keyboards sold



Notes:

(i). The number of keyboards sold by all five shops together is 1200.

(ii). The number of keyboards sold = The number of wired keyboards sold + The number of wireless keyboards sold.

If the ratio of the number of wired keyboards to wireless keyboards sold by Shop B is 5 : 4, find the number of wireless keyboards sold by Shop B.

- 120
- 140
- 160
- 180
- 200

2. Questions

If the number of keyboards sold by shop F is 11.11% more than that by shop C and the number of wireless keyboards sold by shop F is $\frac{1}{3}$ of the number of keyboards sold by shop A. Find the number of wired keyboards sold by shop F.

- 100

- b. 150
- c. 90
- d. 110
- e. 80

3. Questions

Find the ratio between the 25% of the number of keyboards sold by shop A and the 20% of the number of keyboards sold by shop B.

- a. 19 : 18
- b. 25 : 24
- c. 21 : 20
- d. 23 : 22
- e. 18 : 17

4. Questions

The number of keyboards sold by shop C is what percentage more than the number of keyboards sold by shop E?

- a. 20%
- b. 60%
- c. 30%
- d. 40%
- e. 50%

5. Questions

What is the difference between the number of keyboards sold by shops A and D?

- a. 40
- b. 60
- c. 50
- d. 70
- e. 30

6. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the total number of speakers (wired and wireless) sold by five different shops

namely P, Q, R, S and T along with the ratio of the number of wired speakers sold to the number of wireless speakers sold by each shop.

Shop	The total number of speakers sold	The ratio of the number of wired speakers to the wireless speakers sold
P	320	3 : 1
Q	250	2 : 3
R	180	2 : 1
S	240	5 : 3
T	210	2 : 5

Note: The total number of speakers sold = The number of wired speakers sold + The number of wireless speakers sold.

If Shop R sells each wired speaker for Rs. 400 and each wireless speaker for Rs. 600, find the total revenue generated by Shop R.

- a. Rs. 75000
- b. Rs. 68000
- c. Rs. 72000
- d. Rs. 84000
- e. Rs. 96000

7. Questions

If the number of wired speakers sold by shop U is $(\frac{3}{5})$ times more than that sold by shop T and the ratio of the number of wireless speakers sold by shops S to U is 3 : 2, find the total number of speakers sold by shop U.

- a. 156
- b. 132
- c. 148
- d. 165
- e. 124

8. Questions

Find the ratio between the number of wired speakers sold by shop P and the number of wireless speakers sold by shop R.

- a. 2 : 3
- b. 3 : 4
- c. 4 : 1
- d. 5 : 7
- e. 3 : 1

9. Questions

The total number of wired speakers sold by shops Q and R together is how much more or less than the total number of wireless speakers sold by shops P and Q together?

- a. 20 more
- b. 10 less
- c. 10 more
- d. 15 less
- e. 25 more

10. Questions

The number of wired speakers sold by shop S is what percentage of the total number of speakers sold by shop Q?

- a. 30%
- b. 70%
- c. 40%
- d. 50%
- e. 60%

11. Questions

M spends 30% of his monthly income on rent. Out of the remaining income, he spends 40% on groceries and 20% on utility bills and the saves the remaining income. If his monthly savings are Rs. 11200, find his total monthly income.

- a. Rs. 40000
- b. Rs. 30000
- c. Rs. 50000
- d. Rs. 20000

e. Rs. 60000

12. Questions

The speed of a boat in still water is 21 km/hr and the ratio of the downstream speed to the upstream speed of the boat is 5 : 2. Find the time taken by the boat to cover 420 km downstream.

- a. 15 hours
- b. 14 hours
- c. 16 hours
- d. 17 hours
- e. 18 hours

13. Questions

A vessel contains a mixture of juice and water in the ratio of 5 : 2 respectively. If 56 liters of the mixture are taken out and 16 liters of water are added to the vessel, the ratio of juice to water becomes 15 : 8. Find the initial quantity of water (in liters).

- a. 74 liters
- b. 85 liters
- c. 64 liters
- d. 42 liters
- e. 72 liters

14. Questions

Ten friends went to a restaurant for dinner. Nine of them spent Rs. 60 each on their meals, and the tenth person spent Rs. 18 more than the average expenditure of all ten friends. What was the average amount spent by all of them together?

- a. Rs. 60
- b. Rs. 68
- c. Rs. 64
- d. Rs. 62
- e. Rs. 70

15. Questions

The ratio of the cost price to the selling price of an article is 5 : 7. The marked price of the article is Rs. 18000, and it is sold after allowing a discount of 30%. Find the profit earned (in Rs).

- a. Rs 5900

- b. Rs 3200
- c. Rs 4500
- d. Rs 6800
- e. Rs 3600

16. Questions

A woman invested Rs. P in a simple interest scheme at 12% p.a for two years and received an interest of Rs. 1080. If she had invested Rs. $(P + 500)$ in a compound interest scheme (compounded annually) at the same rate of interest for the same period, find the compound interest received by him.

- a. Rs. 1272
- b. Rs. 1380
- c. Rs. 1195
- d. Rs. 1325
- e. Rs. 1280

17. Questions

The age of Vikram two years hence will be half of the present age of Rahul, and the present age of Vikram is 20% less than the present age of Sameer. If Sameer is 22 years younger than Rahul, then find the age of Vikram four years ago.

- a. 18 years
- b. 20 years
- c. 24 years
- d. 22 years
- e. 15 years

18. Questions

Pipe A can fill an empty tank in 24 hours and Pipe B can fill the same tank in 30 hours. Pipe C can empty the completely filled tank in 40 hours. Initially all three pipes are opened together. After 8 hours, Pipe C is closed. Find the time taken by Pipe A and Pipe B to fill the remaining tank.

- a. 7 hours
- b. 6 hours
- c. 8 hours
- d. 9 hours
- e. 5 hours

19. Questions

The length of Train X is 20% less than the length of Train Y, and the speed of Train X is 108 km/hr. If Train X crosses a stationary pole in 16 seconds and overtakes Train Y in 108 seconds while both are running in the same direction, find the time taken by Train Y to cross a 500 m long platform.

- a. 64 seconds
- b. 48 seconds
- c. 24 seconds
- d. 55 seconds
- e. 36 seconds

20. Questions

The area of a circle is 1386 cm^2 . The length of a rectangle is equal to the radius of this circle. If the perimeter of a square is 64 cm and the breadth of the rectangle is 25% less than the side of the square, find the area of the rectangle.

- a. 293 cm^2
- b. 104 cm^2
- c. 264 cm^2
- d. 118 cm^2
- e. 252 cm^2

21. Questions

What value should come in the place of (?) in the following questions.

$$18\% \text{ of } 250 + [(24\% \text{ of } 500) \div 12] = ? - 25$$

- a. 60
- b. 70
- c. 80
- d. 90
- e. 100

22. Questions

$$(2/5) * 350 - \sqrt{289} = ? - (3/8) * 160$$

- a. 163

- b. 173
- c. 203
- d. 193
- e. 183

23. Questions

$$72 \div \sqrt{144} + 14 * 5 = ? + (216)^{1/3}$$

- a. 60
- b. 65
- c. 75
- d. 70
- e. 80

24. Questions

$$15\% \text{ of } 200 + ? - 12\% \text{ of } 50 = \sqrt{[400]\% \text{ of } 150}$$

- a. 4
- b. 6
- c. 2
- d. 3
- e. 5

25. Questions

$$25\% \text{ of } 640 + \sqrt{256} - 30\% \text{ of } 400 = ?$$

- a. 56
- b. 52
- c. 46
- d. 62
- e. 66

26. Questions

What approximate value will come in the place of (?) in the following questions?



$$39.98\% \text{ of } 450.02 + \sqrt{1023.95} - 24.89\% \text{ of } 320.15 = ?$$

- a. 122

- b. 132
- c. 140
- d. 164
- e. 152

27. Questions

$$(14.95)^2 - (11.97)^2 + \sqrt{1520.8} = ? * 4.95$$

- a. 18
- b. 22
- c. 24
- d. 26
- e. 30

28. Questions

$$(15.98)^{2.01} * (63.95)^{1.98} \div 1023.9 = (4)^?$$

- a. 2
- b. 3
- c. 4
- d. 5
- e. 6

29. Questions

$$24.05 * 15.02 - 19.98 * 12.05 = ? * 5.01$$

- a. 24
- b. 60
- c. 48
- d. 8
- e. 32

30. Questions

$$54.98\% \text{ of } 1599.99 - (\sqrt{3135.98} \div 13.99) * 149.95 = ?$$

- a. 320
- b. 260

- c. 240
- d. 300
- e. 280

31. Questions

What value will come in the place of (?) in the following number series?

3, 4, ?, 33, 136, 685, 4116

- a. 10
- b. 8
- c. 5
- d. 9
- e. 12

32. Questions

2, 2.5, 3.5, 5, 7, ?

- a. 7.5
- b. 9.5
- c. 8
- d. 8.5
- e. 9

33. Questions

5, ?, 23, 41, 65, 95

- a. 7
- b. 9
- c. 11
- d. 6
- e. 15

34. Questions

9, 17, 33, 65, 129, ?

- a. 235
- b. 248

- c. 213
- d. 257
- e. 209

35. Questions

17, 18, 27, 52, ?, 182

- a. 111
- b. 72
- c. 109
- d. 83
- e. 101

36. Questions

Find the wrong number in the following number series.

15, 32, 51, 75, 103, 134

- a. 75
- b. 15
- c. 32
- d. 134
- e. 51

37. Questions

10, 13, 22, 47, 96, 217

- a. 13
- b. 10
- c. 22
- d. 47
- e. 217

38. Questions

19, 21.5, 26.5, 35, 44, 56.5

- a. 19
- b. 21.5

- c. 35
- d. 56.5
- e. 44

39. Questions

4, 13, 39, 121, 364, 1093

- a. 121
- b. 13
- c. 4
- d. 39
- e. 1093

40. Questions

8, 25, 59, 110, 180, 263

- a. 8
- b. 110
- c. 59
- d. 25
- e. 180

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 17x + 72 = 0$

II). $y^2 - 15y + 56 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 25x + 156 = 0$

II). $y^2 - 16y + 63 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $2x^2 - 15x + 28 = 0$

II). $2y^2 - 11y + 15 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x > y$
- e. $x \leq y$

44. Questions

I). $x^2 - 47x + 552 = 0$

II). $y^2 - 53y + 696 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.
- d. $x > y$
- e. $x \leq y$

45. Questions

I). $x^2 - 7x + 12 = 0$

II). $y^2 - 8y + 15 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established.

d. $x > y$

e. $x \leq y$

Explanations:

1. Questions

The number of keyboards sold by shop A = $1200 * 25\% = 300$

The number of keyboards sold by shop B = $1200 * 30\% = 360$

The number of keyboards sold by shop C = $1200 * 15\% = 180$

The number of keyboards sold by shop D = $1200 * 20\% = 240$

The number of keyboards sold by shop E = $1200 * 10\% = 120$

Shop	The number of keyboards sold
A	300
B	360
C	180
D	240
E	120

Answer: C

The number of keyboards sold by shop B = 360

The number of wireless keyboards sold by shop B = $360 * \frac{4}{9} = 160$

Hence, the correct answer is option C.

2. Questions

The number of keyboards sold by shop A = $1200 * 25\% = 300$

The number of keyboards sold by shop B = $1200 * 30\% = 360$

The number of keyboards sold by shop C = $1200 * 15\% = 180$

The number of keyboards sold by shop D = $1200 * 20\% = 240$

The number of keyboards sold by shop E = $1200 * 10\% = 120$

Shop	The number of keyboards sold
A	300
B	360
C	180
D	240
E	120

Answer: A

$(11.11\% = 1/9)$

The number of keyboards sold by shop F = $180 * 10/9 = 200$

The number of wireless keyboards sold by shop F = $300 * 1/3 = 100$

The number of wired keyboards sold by shop F = $200 - 100 = 100$

Hence, the correct answer is option A.

3. Questions

The number of keyboards sold by shop A = $1200 * 25\% = 300$

The number of keyboards sold by shop B = $1200 * 30\% = 360$

The number of keyboards sold by shop C = $1200 * 15\% = 180$

The number of keyboards sold by shop D = $1200 * 20\% = 240$

The number of keyboards sold by shop E = $1200 * 10\% = 120$

Shop	The number of keyboards sold
A	300
B	360
C	180
D	240
E	120

Answer: B

The 25% of the number of keyboards sold by shop A = $300 * 25\% = 75$

The 20% of the number of keyboards sold by shop B = $360 * 20\% = 72$

The required ratio = $75 : 72 = 25 : 24$

Hence, the correct answer is option B.

4. Questions

The number of keyboards sold by shop A = $1200 * 25\% = 300$

The number of keyboards sold by shop B = $1200 * 30\% = 360$

The number of keyboards sold by shop C = $1200 * 15\% = 180$

The number of keyboards sold by shop D = $1200 * 20\% = 240$

The number of keyboards sold by shop E = $1200 * 10\% = 120$

Shop	The number of keyboards sold
A	300
B	360
C	180
D	240
E	120

Answer: E

The number of keyboards sold by shop C = 180

The number of keyboards sold by shop E = 120

The required percentage = $(180 - 120) * 100/120 = 50\%$

Hence, the correct answer is option E.

5. Questions

The number of keyboards sold by shop A = $1200 * 25\% = 300$

The number of keyboards sold by shop B = $1200 * 30\% = 360$

The number of keyboards sold by shop C = $1200 * 15\% = 180$

The number of keyboards sold by shop D = $1200 * 20\% = 240$

The number of keyboards sold by shop E = $1200 * 10\% = 120$

Shop	The number of keyboards sold
A	300
B	360
C	180
D	240
E	120

Answer: B

The number of keyboards sold by shop A = 300

The number of keyboards sold by shop D = 240

The required difference = $300 - 240 = 60$

Hence, the correct answer is option B.

6. Questions

Shop P:

The total number of speakers sold by shop P = 320

The number of wired speakers sold by shop P = $320 \times \frac{3}{4} = 240$

The number of wireless speakers sold by shop P = $320 \times \frac{1}{4} = 80$

Shop Q:

The total number of speakers sold by shop Q = 250

The number of wired speakers sold by shop Q = $250 \times \frac{2}{5} = 100$

The number of wireless speakers sold by shop Q = $250 \times \frac{3}{5} = 150$

Shop R:

The total number of speakers sold by shop R = 180

The number of wired speakers sold by shop R = $180 \times \frac{2}{3} = 120$

The number of wireless speakers sold by shop R = $180 \times \frac{1}{3} = 60$

Shop S:

The total number of speakers sold by shop S = 240

The number of wired speakers sold by shop S = $240 \times \frac{5}{8} = 150$

The number of wireless speakers sold by shop S = $240 \times \frac{3}{8} = 90$

Shop T:

The total number of speakers sold by shop T = 210

The number of wired speakers sold by shop T = $210 \times \frac{2}{7} = 60$

The number of wireless speakers sold by shop T = $210 \times \frac{5}{7} = 150$

Shop	The number of wired speakers sold	The number of wireless speakers sold	The total number of speakers sold
P	240	80	320
Q	100	150	250
R	120	60	180
S	150	90	240
T	60	150	210

Answer: D

The number of wired speakers sold by shop R = 120

The revenue from selling wired speakers = $120 \times 400 = \text{Rs. } 48000$

The number of wireless speakers sold by shop R = 60

The revenue from selling wireless speakers = $60 \times 600 = \text{Rs. } 36000$

The total revenue = $48000 + 36000 = \text{Rs. } 84000$

Hence, the correct answer is option D.

7. Questions

Shop P:

The total number of speakers sold by shop P = 320

The number of wired speakers sold by shop P = $320 \times \frac{3}{4} = 240$

The number of wireless speakers sold by shop P = $320 \times \frac{1}{4} = 80$

Shop Q:

The total number of speakers sold by shop Q = 250

The number of wired speakers sold by shop Q = $250 \times \frac{2}{5} = 100$

The number of wireless speakers sold by shop Q = $250 \times \frac{3}{5} = 150$

Shop R:

The total number of speakers sold by shop R = 180

The number of wired speakers sold by shop R = $180 * \frac{2}{3} = 120$

The number of wireless speakers sold by shop R = $180 * \frac{1}{3} = 60$

Shop S:

The total number of speakers sold by shop S = 240

The number of wired speakers sold by shop S = $240 * \frac{5}{8} = 150$

The number of wireless speakers sold by shop S = $240 * \frac{3}{8} = 90$

Shop T:

The total number of speakers sold by shop T = 210

The number of wired speakers sold by shop T = $210 * \frac{2}{7} = 60$

The number of wireless speakers sold by shop T = $210 * \frac{5}{7} = 150$

Shop	The number of wired speakers sold	The number of wireless speakers sold	The total number of speakers sold
P	240	80	320
Q	100	150	250
R	120	60	180
S	150	90	240
T	60	150	210

Answer: A

The number of wired speakers sold by shop T = 60

The number of wired speakers sold by shop U = $60 * \frac{8}{5} = 96$

The number of wireless speakers sold by shop U = $90 * \frac{2}{3} = 60$

The total number of speakers sold by shop U = $96 + 60 = 156$

Hence, the correct answer is option A.

8. Questions

Shop P:

The total number of speakers sold by shop P = 320

The number of wired speakers sold by shop P = $320 * \frac{3}{4} = 240$

The number of wireless speakers sold by shop P = $320 * \frac{1}{4} = 80$

Shop Q:

The total number of speakers sold by shop Q = 250

The number of wired speakers sold by shop Q = $250 \times \frac{2}{5} = 100$

The number of wireless speakers sold by shop Q = $250 \times \frac{3}{5} = 150$

Shop R:

The total number of speakers sold by shop R = 180

The number of wired speakers sold by shop R = $180 \times \frac{2}{3} = 120$

The number of wireless speakers sold by shop R = $180 \times \frac{1}{3} = 60$

Shop S:

The total number of speakers sold by shop S = 240

The number of wired speakers sold by shop S = $240 \times \frac{5}{8} = 150$

The number of wireless speakers sold by shop S = $240 \times \frac{3}{8} = 90$

Shop T:

The total number of speakers sold by shop T = 210

The number of wired speakers sold by shop T = $210 \times \frac{2}{7} = 60$

The number of wireless speakers sold by shop T = $210 \times \frac{5}{7} = 150$

Shop	The number of wired speakers sold	The number of wireless speakers sold	The total number of speakers sold
P	240	80	320
Q	100	150	250
R	120	60	180
S	150	90	240
T	60	150	210

Answer: C

The number of wired speakers sold by shop P = 240

The number of wireless speakers sold by shop R = 60

The required ratio = $240 : 60 = 4 : 1$

Hence, the correct answer is option C.

9. Questions

Shop P:

The total number of speakers sold by shop P = 320

The number of wired speakers sold by shop P = $320 \times \frac{3}{4} = 240$

The number of wireless speakers sold by shop P = $320 \times \frac{1}{4} = 80$

Shop Q:

The total number of speakers sold by shop Q = 250

The number of wired speakers sold by shop Q = $250 \times \frac{2}{5} = 100$

The number of wireless speakers sold by shop Q = $250 \times \frac{3}{5} = 150$

Shop R:

The total number of speakers sold by shop R = 180

The number of wired speakers sold by shop R = $180 \times \frac{2}{3} = 120$

The number of wireless speakers sold by shop R = $180 \times \frac{1}{3} = 60$

Shop S:

The total number of speakers sold by shop S = 240

The number of wired speakers sold by shop S = $240 \times \frac{5}{8} = 150$

The number of wireless speakers sold by shop S = $240 \times \frac{3}{8} = 90$

Shop T:

The total number of speakers sold by shop T = 210

The number of wired speakers sold by shop T = $210 \times \frac{2}{7} = 60$

The number of wireless speakers sold by shop T = $210 \times \frac{5}{7} = 150$

Shop	The number of wired speakers sold	The number of wireless speakers sold	The total number of speakers sold
P	240	80	320
Q	100	150	250
R	120	60	180
S	150	90	240
T	60	150	210

Answer: B

The total number of wired speakers sold by shops Q and R together = $100 + 120 = 220$

The total number of wireless speakers sold by shops P and Q together = $80 + 150 = 230$

The required answer = $230 - 220 = 10$ less

Hence, the correct answer is option B.

10. Questions

Shop P:

The total number of speakers sold by shop P = 320

The number of wired speakers sold by shop P = $320 * \frac{3}{4} = 240$

The number of wireless speakers sold by shop P = $320 * \frac{1}{4} = 80$

Shop Q:

The total number of speakers sold by shop Q = 250

The number of wired speakers sold by shop Q = $250 * \frac{2}{5} = 100$

The number of wireless speakers sold by shop Q = $250 * \frac{3}{5} = 150$

Shop R:

The total number of speakers sold by shop R = 180

The number of wired speakers sold by shop R = $180 * \frac{2}{3} = 120$

The number of wireless speakers sold by shop R = $180 * \frac{1}{3} = 60$

Shop S:

The total number of speakers sold by shop S = 240

The number of wired speakers sold by shop S = $240 * \frac{5}{8} = 150$

The number of wireless speakers sold by shop S = $240 * \frac{3}{8} = 90$

Shop T:

The total number of speakers sold by shop T = 210

The number of wired speakers sold by shop T = $210 * \frac{2}{7} = 60$

The number of wireless speakers sold by shop T = $210 * \frac{5}{7} = 150$

Shop	The number of wired speakers sold	The number of wireless speakers sold	The total number of speakers sold
P	240	80	320
Q	100	150	250
R	120	60	180
S	150	90	240
T	60	150	210

Answer: E

The number of wired speakers sold by shop S = 150

The total number of speakers sold by shop Q = 250

The required percentage = $150 \times 100/250 = 60\%$

Hence, the correct answer is option E.

11. Questions

Answer: A

Let the total monthly income of M = 100a

The amount spent on rent = $100a \times 30\% = 30a$

The remaining amount = 70a

The amount spent on groceries and utility bills = $(40 + 20)\% \times 70a = 42a$

The remaining amount = $70a - 42a = 28a$

According to the question,

$$28a = 11200$$

$$a = 400$$

The monthly income of M = $100 \times 400 = \text{Rs. } 40000$

Hence, the correct answer is option A.

12. Questions

Answer: B

The speed of the boat in still water = 21 km/hr

Let the speed of the stream = x km/hr

According to the question,

$$(21 + x) / (21 - x) = 5 / 2$$

$$42 + 2x = 105 - 5x$$

$$7x = 63$$

$$x = 9$$

The downstream speed of the boat = $21 + 9 = 30$ km/hr

The required time = $420/30 = 14$ hours

Hence, the correct answer is option B.

13. Questions

Answer: C

Let the initial quantity of the juice = $5a$

The initial quantity of water = $2a$

The removed quantity of juice = $56 * 5/7 = 40$ litres

The removed quantity of water = $56 - 40 = 16$ litres

According to the question,

$$(5a - 40) / (2a - 16 + 16) = 15 / 8$$

$$40a - 320 = 30a$$

$$10a = 320$$

$$a = 32$$

The initial quantity of water = $32 * 2 = 64$ litres

Hence, the correct answer is option C.

14. Questions

Answer: D

Let the average amount spent by all ten = x

The total amount spent by all ten = $10x$

According to the question,

$$60 * 9 + (x + 18) = 10x$$

$$540 + x + 18 = 10x$$

$$558 = 9x$$

$$x = 62$$

The required answer = Rs. 62

Hence, the correct answer is option D.

15. Questions

Answer: E

The marked price of the article = Rs. 18000

The selling price of the article = Rs. 18000 * 70% = Rs. 12600

The cost price of the article = 12600 * 5/7 = Rs. 9000

The required difference = 12600 - 9000 = Rs. 3600

Hence, the correct answer is option E.

16. Questions

Answer: A

The simple interest = $PRT/100$

$$1080 = P * 12\% * 2$$

$$P = 4500$$

The amount invested for compound interest = 4500 + 500 = Rs. 5000

The effective rate of compound interest = $12 + 12 + 144/100 = 25.44\%$

The compound interest = 5000 * 25.44% = Rs. 1272

Hence, the correct answer is option A.

17. Questions

Answer: B

$$(20\% = 1/5)$$

Let the present age of Sameer = 5a

The present age of Vikram = 4a

The present age of Rahul = $(4a + 2) * 2 = 8a + 4$

According to the question,

$$8a + 4 - 5a = 22$$

$$3a = 18$$

$$a = 6$$

The present age of Vikram = $4 * 6 = 24$ years

The age of Vikram 4 years ago = $24 - 4 = 20$ years

Hence, the correct answer is option B.

18. Questions

Answer: C

Let the total capacity of the tank = 120 units [LCM of 24, 30, 40]

The efficiency of pipe A to fill the tank = $120 / 24 = 5$ units per hour

The efficiency of pipe B to fill the tank = $120 / 30 = 4$ units per hour

The efficiency of pipe C to empty the tank = $120 / 40 = 3$ units per hour

The capacity of the tank filled by all three pipes in 8 hours = $(4 + 5 - 3) * 8 = 48$ units.

The remaining capacity of the tank = $120 - 48 = 72$ units

The time taken by pipe A and B to fill the remaining tank = $72 / 9 = 8$ hours

Hence, the correct answer is option C.

19. Questions

Answer: D

Let the length of the train Y = $5a$

The length of the train X = $4a$

The speed of the train X = $108 * 5 / 18 = 30$ m/sec

According to the question,

$$4a / 30 = 16$$

$$a = 120$$

The length of the train Y = $5a = 5 * 120 = 600$ m

Let the speed of the train Y = b

The relative speed of the train X and Y in same direction = $30 - b$

The total distance travelled by train X = $480 + 600 = 1080$ m

According to the question,

$$30 - b = 1080 / 108$$

$$30 - b = 10$$

$$b = 20 \text{ m/sec}$$

The required time = $(600 + 500) / 20 = 55$ seconds

Hence, the correct answer is option D.

20. Questions

Answer: E

The perimeter of the square = $4 * \text{side of square}$

The side of square = $64 / 4 = 16$ cm

The breadth of the rectangle = $16 \times \frac{3}{4} = 12$ cm

The area of the circle = πr^2

So,

$$(22/7) \times r^2 = 1386$$

$$r^2 = 441$$

$$r = 21 \text{ cm}$$

The length of the rectangle = 21 cm

The area of the rectangle = length * breadth = $21 \times 12 = 252 \text{ cm}^2$

Hence, the correct answer is option E.

21. Questions

Answer: C

$$18\% \text{ of } 250 + [(24\% \text{ of } 500) \div 12] = ? - 25$$

$$45 + [120 \div 12] = ? - 25$$

$$45 + 10 = ? - 25$$

$$55 = ? - 25$$

$$55 + 25 = ?$$

$$80 = ?$$

Hence, the correct answer is option C.

22. Questions

Answer: E

$$(2/5) \times 350 - \sqrt{289} = ? - (3/8) \times 160$$

$$2 \times 70 - 17 = ? - 3 \times 20$$

$$140 - 17 = ? - 60$$

$$123 = ? - 60$$

$$123 + 60 = ?$$

$$183 = ?$$

Hence, the correct answer is option E.

23. Questions

Answer: D

$$72 \div \sqrt{144} + 14 \times 5 = ? + (216)^{1/3}$$

$$72 \div 12 + 70 = ? + 6$$

$$6 + 70 = ? + 6$$

$$76 = ? + 6$$

$$70 = ?$$

Hence, the correct answer is option D.

24. Questions

Answer: B

$$15\% \text{ of } 200 + ? - 12\% \text{ of } 50 = \sqrt{[400]\% \text{ of } 150}$$

$$30 + ? - 6 = 20/100 * 150$$

$$24 + ? = 30$$

$$? = 30 - 24$$

$$? = 6$$

Hence, the correct answer is option B.

25. Questions

Answer: A

$$25\% \text{ of } 640 + \sqrt{256} - 30\% \text{ of } 400 = ?$$

$$160 + 16 - 120 = ?$$

$$176 - 120 = ?$$

$$56 = ?$$

Hence, the correct answer is option A.

26. Questions

Answer: B

$$39.98\% \text{ of } 450.02 + \sqrt{1023.95} - 24.89\% \text{ of } 320.15 = ?$$

$$40\% \text{ of } 450 + \sqrt{1024} - 25\% \text{ of } 320 = ?$$

$$180 + 32 - 80 = ?$$

$$212 - 80 = ?$$

$$132 = ?$$

Hence, the correct answer is option B.

27. Questions

Answer: C

$$(14.95)^2 - (11.97)^2 + \sqrt{1520.8} = ? * 4.95$$

$$(15)^2 - (12)^2 + \sqrt{1521} = ? * 5$$

$$225 - 144 + 39 = ? * 5$$

$$81 + 39 = ? * 5$$

$$120 = ? * 5$$

$$? = 24$$

Hence, the correct answer is option C.

28. Questions

Answer: D

$$(15.98)^{2.01} * (63.95)^{1.98} \div 1023.9 = (4)^?$$

$$(16)^2 * (64)^2 \div 1024 = (4)^?$$

$$((4)^2)^2 * ((4)^3)^2 \div (4)^5 = (4)^?$$

$$(4)^4 * (4)^6 \div (4)^5 = (4)^?$$

$$(4)^{(4+6-5)} = (4)^?$$

$$(4)^5 = (4)^?$$

$$? = 5$$

Hence, the correct answer is option D.

29. Questions

Answer: A

$$24.05 * 15.02 - 19.98 * 12.05 = ? * 5.01$$

$$24 * 15 - 20 * 12 = ? * 5$$

$$360 - 240 = ? * 5$$

$$120 = ? * 5$$

$$? = 24$$

Hence, the correct answer is option A.

30. Questions

Answer: E

$$54.98\% \text{ of } 1599.99 - (\sqrt{3135.98} \div 13.99) * 149.95 = ?$$

$$55\% \text{ of } 1600 - (\sqrt{3136} \div 14) * 150 = ?$$

$$880 - (56 \div 14) * 150 = ?$$

$$880 - 4 * 150 = ?$$

$$880 - 600 = ?$$

$$280 = ?$$

Hence, the correct answer is option E.

31. Questions

Answer: A

$$3 * 1 + 1 = 4$$

$$4 * 2 + 2 = 10$$

$$10 * 3 + 3 = \mathbf{33}$$

$$33 * 4 + 4 = 136$$

$$136 * 5 + 5 = 685$$

$$685 * 6 + 6 = 4116$$

Missing number: 10

Hence, the correct answer is option A.

32. Questions

Answer: B

$$2 + 0.5 = 2.5$$

$$2.5 + 1 = 3.5$$

$$3.5 + 1.5 = 5$$

$$5 + 2 = 7$$

$$7 + 2.5 = \mathbf{9.5}$$

Missing number: 9.5

Hence, the correct answer is option B.

33. Questions

Answer: C

$$5 + 6 = \mathbf{11}$$

$$11 + 12 = 23$$

$$23 + 18 = 41$$

$$41 + 24 = 65$$

$$65 + 30 = 95$$

Missing number: 11

Hence, the correct answer is option C.

34. Questions

Answer: D

$$9 * 2 - 1 = 17$$

$$17 * 2 - 1 = 33$$

$$33 * 2 - 1 = 65$$

$$65 * 2 - 1 = 129$$

$$129 * 2 - 1 = \mathbf{257}$$

Missing number: 257

Hence, the correct answer is option D.

35. Questions

Answer: E

$$17 + 1^2 = 18$$

$$18 + 3^2 = 27$$

$$27 + 5^2 = 52$$

$$52 + 7^2 = \mathbf{101}$$

$$101 + 9^2 = 182$$

Missing number: 101

Hence, the correct answer is option E.

36. Questions

Answer: A

$$15 + 17 = 32$$

$$32 + 19 = 51$$

$$51 + 23 = \mathbf{74}$$

$$74 + 29 = 103$$

$$103 + 31 = 134$$

Wrong number: 75

Right number: 74

Hence, the correct answer is option A.

37. Questions

Answer: B

$$9 + 2^2 = 13$$

$$13 + 3^2 = 22$$

$$22 + 5^2 = 47$$

$$47 + 7^2 = 96$$

$$96 + 11^2 = 217$$

Wrong number: 10

Right number: 9

Hence, the correct answer is option B.

38. Questions

Answer: C

$$19 + 2.5 = 21.5$$

$$21.5 + 5 = 26.5$$

$$26.5 + 7.5 = 34$$

$$34 + 10 = 44$$

$$44 + 12.5 = 56.5$$

Wrong number: 35

Right number: 34

Hence, the correct answer is option C.

39. Questions

Answer: D

$$4 * 3 + 1 = 13$$

$$13 * 3 + 1 = 40$$

$$40 * 3 + 1 = 121$$

$$121 * 3 + 1 = 364$$

$$364 * 3 + 1 = 1093$$

Wrong number: 39

Right number: 40

Hence, the correct answer is option D.

40. Questions

Answer: E

$$8 + 17 = 25$$

$$25 + 34 = 59$$

$$59 + 51 = 110$$

$$110 + 68 = \mathbf{178}$$

$$178 + 85 = 263$$

Wrong number: 180

Right number: 178

Hence, the correct answer is option E.

41. Questions

Answer: B

I). $x^2 - 17x + 72 = 0$

$$x^2 - 9x - 8x + 72 = 0$$

$$x(x - 9) - 8(x - 9) = 0$$

$$(x - 8)(x - 9) = 0$$

$$x = 8 \text{ or } 9$$

II). $y^2 - 15y + 56 = 0$

$$y^2 - 8y - 7y + 56 = 0$$

$$y(y - 8) - 7(y - 8) = 0$$

$$(y - 7)(y - 8) = 0$$

$$y = 7 \text{ or } 8$$

So, $x \geq y$

Hence, the correct answer is option B.

42. Questions

Answer: A

I). $x^2 - 25x + 156 = 0$

$$x^2 - 12x - 13x + 156 = 0$$

$$x(x - 12) - 13(x - 12) = 0$$

$$(x - 13)(x - 12) = 0$$

$$x = 13 \text{ or } 12$$

II). $y^2 - 16y + 63 = 0$

$$y^2 - 9y - 7y + 63 = 0$$

$$y(y - 9) - 7(y - 9) = 0$$

$$(y - 7)(y - 9) = 0$$

$$y = 7 \text{ or } 9$$

So, $x > y$

Hence, the correct answer is option A.

43. Questions**Answer: D**

I). $2x^2 - 15x + 28 = 0$

$$2x^2 - 8x - 7x + 28 = 0$$

$$2x(x - 4) - 7(x - 4) = 0$$

$$(2x - 7)(x - 4) = 0$$

$$x = 7/2 \text{ or } 4$$

II). $2y^2 - 11y + 15 = 0$

$$2y^2 - 6y - 5y + 15 = 0$$

$$2y(y - 3) - 5(y - 3) = 0$$

$$(2y - 5)(y - 3) = 0$$

$$y = 5/2 \text{ or } 3$$

So, $x > y$

Hence, the correct answer is option D.

44. Questions**Answer: E**

I). $x^2 - 47x + 552 = 0$

$$x^2 - 24x - 23x + 552 = 0$$

$$x(x - 24) - 23(x - 24) = 0$$

$$(x - 24)(x - 23) = 0$$

$$x = 24 \text{ or } 23$$

II). $y^2 - 53y + 696 = 0$

$$y^2 - 29y - 24y + 696 = 0$$

$$y(y - 29) - 24(y - 29) = 0$$

$$(y - 24)(y - 29) = 0$$

$$y = 24 \text{ or } 29$$

So, $x \leq y$

Hence, the correct answer is option E.

45. Questions

Answer: C

I). $x^2 - 7x + 12 = 0$

$$x^2 - 4x - 3x + 12 = 0$$

$$x(x - 4) - 3(x - 4) = 0$$

$$(x - 3)(x - 4) = 0$$

$$x = 3 \text{ or } 4$$

II). $y^2 - 8y + 15 = 0$

$$y^2 - 5y - 3y + 15 = 0$$

$$y(y - 5) - 3(y - 5) = 0$$

$$(y - 5)(y - 3) = 0$$

$$y = 5 \text{ or } 3$$

So, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C.